Vidyalankar Institute of Technology Accredited A+ by NAAC	Department of Computer Engineering A.Y-2026-27

Semester	B.E. Semester VII – CMPN
Subject	DevOps
Subject Professor In-charge	Prof. Snehal Andhare
Assisting Teachers	Prof. Snehal Andhare
Laboratory	

Student Name	Vaishnavi Khopkar	
Roll Number	23102A0074	
Grade and Subject Teacher's Signature		

Experiment Number	15
Experiment Title	Luggage Complaint Portal: Final End-to-End Release, Documentation and Viva
Objectives (Skill Set / Knowledge Tested / Imparted)	To run the complete DevOps workflow from Git commit to Jenkins build, Selenium quality gate, Docker deployment and Ansible provisioning, and to complete the documentation.
Program	**1. End-to-end workflow**

Stage	Tool	Result
Commit and push	Git, GitHub	Code on the develop branch, release tagged v1.0.0
Build and unit tests	Jenkins, Maven, JUnit	22 unit tests passing
Quality gate	Selenium	4 journeys passing; failed tests stop deployment
Deploy	Tomcat	Application on port 8082
Package and publish	Docker, local registry	Versioned images (tags 7, 8, 10, latest)
Run	Docker container	Application on port 8083
Provision	Ansible	Configured node with health check and rollback

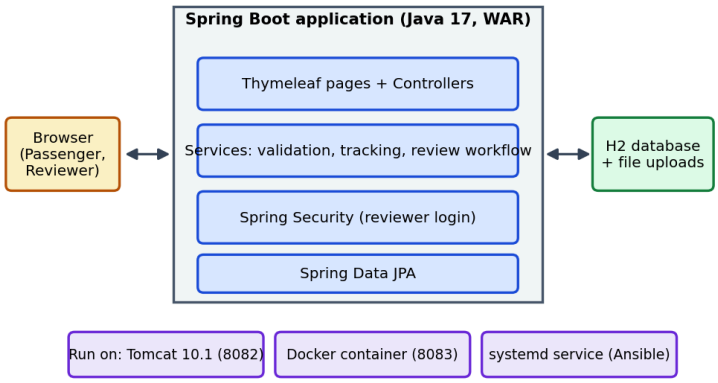


Figure: Architecture of the portal and its deployment targets

2. Final pipeline run

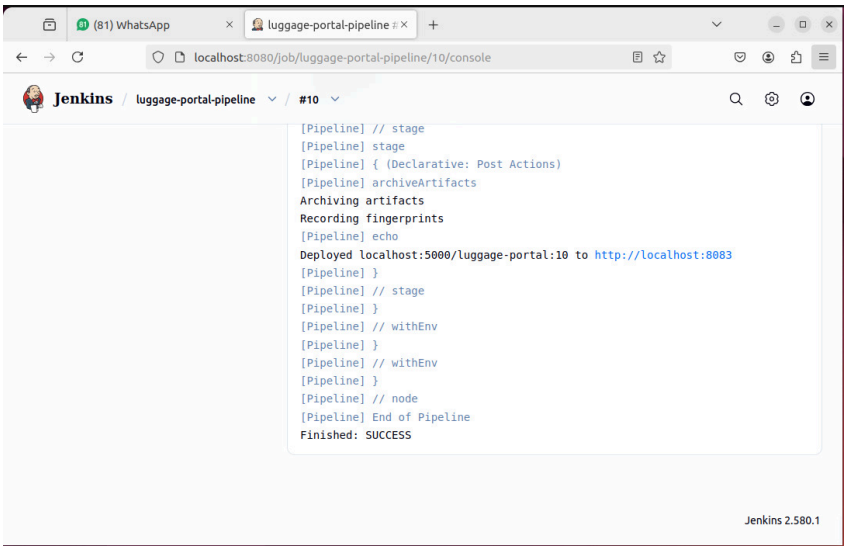


Figure: Build #10: commit to container, Finished: SUCCESS

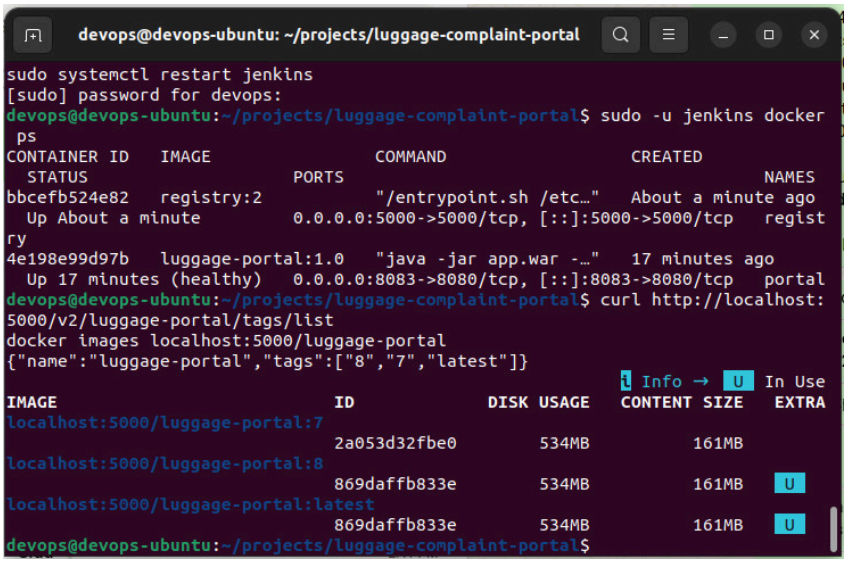


Figure: Versioned images in the registry and the running container

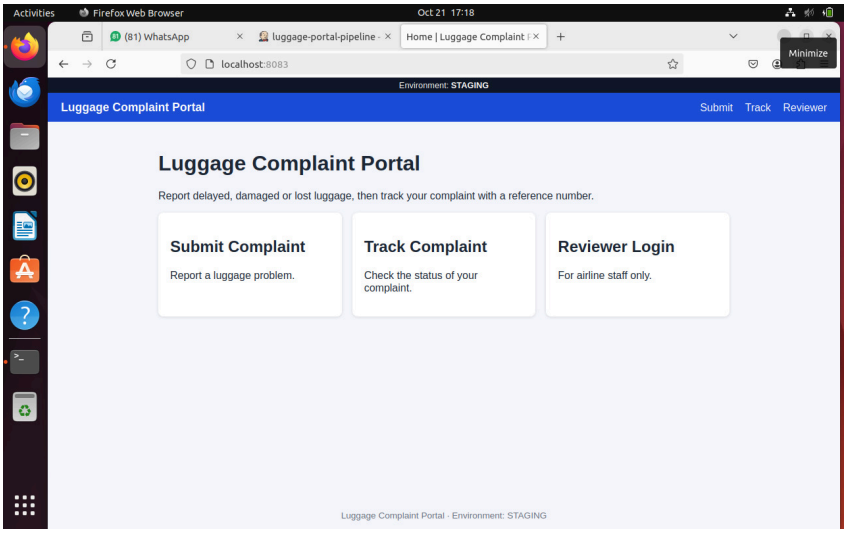


Figure: Final application on port 8083

3. Troubleshooting guide

Problem	Cause	Fix
release version 17 not supported	Default Java was a Java 21 runtime without compiler	Set JAVA_HOME to Java 17
Could not resolve host	VM DNS dropped	Set DNS 8.8.8.8 and 1.1.1.1
Jenkins archive found nothing	Project builds a WAR, not a JAR	Archive target/*.war
Permission denied in /opt/tomcat	Folder owned by the tomcat user	Add jenkins to the tomcat group, chmod 755
Docker permission denied	User not in the docker group	usermod -aG docker, restart Jenkins

Port already in use	Several services on one VM	Fixed port plan: 8080, 8081, 8082, 8083, 8090, 5000
apt NO_PUBKEY (Ansible)	Leftover GitHub CLI repository	Remove the repository file

4. Limitations

- H2 embedded database: data is not suitable for production.
- Everything runs on a single VM.
- Local registry without TLS or authentication.
- Demo reviewer account and no real airline integration.
- Builds are started by polling or manually, not by webhooks.

5. Future enhancement plan

- PostgreSQL with persistent storage and backups.
- Email and SMS notifications to passengers.
- GitHub webhooks and Jenkins agents on separate nodes.
- Kubernetes deployment, HTTPS and secrets management.
- Monitoring and alerting (Prometheus and Grafana).

[Screenshot to be added: Ansible provisioning run, live demonstration video and viva presentation]

Conclusion	The complete workflow ran from a commit through build, test, a versioned Docker image and a fresh container. Troubleshooting notes, limitations and a future enhancement plan are documented.